08.07 Coffee Time Clock Hands

Date Object

forEach()

var clock = document.querySelector('.clock'); var sunArmCoffeeCup = document.querySelector('.cup-arm'); var degRotate = -20; var degCounter = 0; sunArmCoffeeCup.style.transform = 'rotate(' + degRotate + 'deg)';

setInterval(() => {

var date = new Date(); var hour = date.getHours(); // 0-23 var mins = date.getMinutes(); // 0-59 var msec = date.getTime(); // milliseconds since 1970 -- a huge number! // The getTime() method returns the number of milliseconds since the ECMAScript epoch // milliseconds since Jan 1, 1970, 00:00:00.000 GMT // milliseconds / 60 = seconds for Rolex-like smooth sweeping second hand

console.log('msec:', msec); // "normalize" time units so that they are all in 0-1 range, with 1 being a // full 360 turn on the clock hand; so when hour = 6, hour for purposes, // of positioning the hour hand on the clock, needs to be normalized to 0.5 // on top of that, hour needs to have the normalized minutes, divided by 12, // added to it // for hour, we need to only have values from 0-12.. so for 12-23, subtract 12 // since we have to do math to arrive at norm values, and since conditions // apply as well, start by just initializing all norms to 0: var hourNorm = 0; var minsNorm = 0; var secsNorm = msec;

// adjust the hour from mil time to am-pm (0-23 to 0-11) if (hour >= 12) { hour -= 12; } // calc normalized hour (0-12 => from 0-1), so divide by 12.. later we will add // hourMin/12 to hourNorm hourNorm = hour / 12 - 0.05; // normalize hour hand and add 'bias'

secsNorm = msec / 1000; // seconds since 1970 -- but as float secsNorm /= 60; // divide that by 60 to get "normalized multiple" (remainder after doing x turns around clock is in 0-1 range)

// add minNorm fraction to hourNorm, so that at, for example, 7:30, // the hour hand is half-way between 7 and 8 minsNorm = mins / 60; // divide mins by 60 to normalize (get 0-1 range)

hourNorm += minsNorm / 6; // divide by 6 results in more accurate hour hand position than dividing by 12

minsNorm = mins / 60; // tweak the min hand

// same for minsNorm? -- min hand should move w sec hand OR just tick every min

var timeUnits = [hourNorm, minsNorm, secsNorm];

// console.log(timeUnits[0] + ':' + timeUnits[1] + ':' + timeUnits[2]);

// run a func on each item in positions arr where the current array item // is timeUnit and arrIndex is the index of same // the one line of code targets the child of the clock div: // the 3 children of the clock div are hr, min and sec hands, so // as the time unit is looped through, that unit is applied as the rotation // turn value of the corresponding clock hand div // 1 turn = 360 degrees timeUnits.forEach((timeunit, arrindex) => { clock.children[arrindex].style.transform = 'rotate(' + timeunit + 'turn)'; });

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degCounter++; // every 1 min, reset degree counter to 0 if (degCounter >= 2000) { // 40 times per sec x 60 sec = 2400 per min degCounter = 0; }

// rotate sun arm coffee cup -- rotation reg pt is the shoulder if (degCounter < 500 || degCounter > 1500) { degRotate += 0.075; // 1/10 of a degree, so 40x per sec rotates 1/20 deg // so, in 1 sec rotates 4 degs } else { degRotate -= 0.075; }

sunArmCoffeeCup.style.transform = 'rotate(' + degRotate + 'deg)';

}, 40);

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